

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,572
held-out plate appearances

3.03%
held-out HR rate

0.6007
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-28, scored on the 32,572 that came after (actual HR rate 3.03%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5853	1.56×	0.02935	0.13493	0.00452
KNN comps Prior is the pooled rate of the nearest comparables.	0.6007	1.51×	0.02932	0.13397	0.00417
SVR Support-vector regression on the same feature matrix.	0.5980	1.50×	0.02935	0.13448	0.00604
Random forest Forest regression; a learned similarity metric.	0.5972	1.45×	0.02935	0.13433	0.00596
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6007	1.51×	0.02932	0.13397	0.00417
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6007	1.51×	0.02932	0.13397	0.00417
Linear regression Ordinary least squares on the same features.	0.5969	1.50×	0.02933	0.13427	0.00421
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5978	1.54×	0.02935	0.13440	0.00622
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.53×	0.02934	0.13427	0.00486
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5973	1.46×	0.02935	0.13439	0.00611

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,530 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5569	1.45×	0.02299	0.11186	0.01009
KNN comps Prior is the pooled rate of the nearest comparables.	0.5834	1.33×	0.02302	0.11167	0.01228
SVR Support-vector regression on the same feature matrix.	0.5836	1.33×	0.02316	0.11456	0.01754
Random forest Forest regression; a learned similarity metric.	0.5925	1.08×	0.02317	0.11288	0.01708
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5834	1.33×	0.02302	0.11167	0.01228
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5834	1.33×	0.02302	0.11167	0.01228
Linear regression Ordinary least squares on the same features.	0.5861	1.08×	0.02310	0.11311	0.01521
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	1.08×	0.02311	0.11262	0.01479
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5854	1.20×	0.02306	0.11227	0.01442
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6004	1.20×	0.02318	0.11289	0.01705

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Munetaka Murakami 15.7%, Randal Grichuk 13.9%, Miguel Vargas 13.9%, Jo Adell 12.9%, Kyle Teel 11.6%, David Fry 10.1%
Flat league	5/6	Munetaka Murakami 14.0%, Randal Grichuk 12.6%, Jo Adell 12.3%, Miguel Vargas 12.2%, Kyle Teel 11.1%, Angel Martínez 9.8%
KNN comps	6/6	Munetaka Murakami 15.7%, Randal Grichuk 13.7%, Miguel Vargas 13.7%, Jo Adell 12.7%, Kyle Teel 11.7%, David Fry 10.1%
SVR	6/6	Munetaka Murakami 16.3%, Randal Grichuk 14.5%, Miguel Vargas 13.8%, Jo Adell 13.2%, Kyle Teel 10.9%, David Fry 10.1%
Random forest	6/6	Munetaka Murakami 16.0%, Randal Grichuk 13.7%, Miguel Vargas 13.6%, Jo Adell 13.0%, Kyle Teel 11.4%, David Fry 9.9%
Core (KNN+SVR)	6/6	Munetaka Murakami 15.7%, Randal Grichuk 13.7%, Miguel Vargas 13.7%, Jo Adell 12.7%, Kyle Teel 11.7%, David Fry 10.1%
Forest blend	6/6	Munetaka Murakami 15.7%, Randal Grichuk 13.7%, Miguel Vargas 13.7%, Jo Adell 12.7%, Kyle Teel 11.7%, David Fry 10.1%
Linear regression	6/6	Munetaka Murakami 15.5%, Miguel Vargas 13.8%, Randal Grichuk 13.6%, Jo Adell 12.9%, Kyle Teel 11.4%, David Fry 9.9%
Logistic regression	5/6	Munetaka Murakami 15.8%, Randal Grichuk 14.5%, Miguel Vargas 14.0%, Jo Adell 13.2%, Kyle Teel 10.5%, Colson Montgomery 10.4%
Neural net (torch)	5/6	Munetaka Murakami 15.8%, Randal Grichuk 14.2%, Miguel Vargas 13.9%, Jo Adell 13.2%, Kyle Teel 11.0%, Colson Montgomery 9.8%
XGBoost	6/6	Munetaka Murakami 16.2%, Randal Grichuk 13.9%, Miguel Vargas 13.7%, Jo Adell 13.1%, Kyle Teel 11.3%, David Fry 9.7%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T
1Munetaka Murakami	slate15.7%	14.0%	15.7%	16.3%	16.0%	15.7%	15.7%	15.5%	15.8%	
2Randal Grichuk	CWS13.9%	12.6%	13.7%	14.5%	13.7%	13.7%	13.7%	13.6%	14.5%	
3Miguel Vargas	CWS13.9%	12.2%	13.7%	13.8%	13.6%	13.7%	13.7%	13.8%	14.0%	
4Jo Adell	LEslate12.9%	12.3%	12.7%	13.2%	13.0%	12.7%	12.7%	12.9%	13.2%	
5Kyle Teel	CWS11.6%	11.1%	11.7%	10.9%	11.4%	11.7%	11.7%	11.4%	10.5%	
6David Fry	CLE10.1%	9.5%	10.1%	10.1%	9.9%	10.1%	10.1%	9.9%	9.3%	
7Angel Martínez	CLE9.7%	9.8%	9.6%	9.8%	9.6%	9.6%	9.6%	9.6%	9.7%	
8Nathaniel Lowe	CLE9.6%	9.1%	9.5%	9.3%	9.8%	9.5%	9.5%	9.3%	9.3%	
9Colson Montgomery	CWS9.5%	8.5%	9.6%	9.2%	9.6%	9.6%	9.6%	9.9%	10.4%	
10Chase DeLauter	CLE9.4%	9.5%	9.4%	9.2%	9.2%	9.4%	9.4%	9.2%	9.1%	
11Patrick Bailey	CLE9.1%	8.6%	9.1%	8.8%	8.9%	9.1%	9.1%	9.0%	8.5%	
12Tristan Peters	CWS8.2%	8.8%	8.3%	8.7%	8.7%	8.3%	8.3%	8.5%	8.5%	
13Travis Bazzana	CLE8.0%	8.1%	7.9%	7.7%	8.2%	7.9%	7.9%	7.9%	7.6%	
14Chase Meidroth	CWS7.6%	8.7%	7.8%	8.0%	8.0%	7.8%	7.8%	7.9%	7.7%	
15José Ramírez	CLE7.6%	8.0%	7.8%	7.3%	7.8%	7.8%	7.8%	7.9%	8.2%	
16Sam Antonacci	CWS7.3%	8.9%	7.4%	6.6%	7.5%	7.4%	7.4%	7.4%	7.2%	
17Braden Montgomery	CWS5.9%	6.8%	5.6%	5.3%	5.7%	5.6%	5.6%	5.5%	5.4%	
18Bryan Rocchio	CLE5.7%	6.2%	5.6%	5.8%	5.3%	5.6%	5.6%	5.5%	5.5%	
19Steven Kwan	CLE3.6%	6.6%	4.5%	3.5%	2.8%	4.5%	4.5%	2.2%	3.7%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.